

Non logical symbols - Example

Non logical symbols depend on the domain we want to model. They should have an intended meaning in such a domain.

Example (Domain of arithmetics)

symbol	type	arity	intended meaning
0	constant	0*	the smallest natural number
$succ(\cdot)$	function	1	the function that given a number returns its successor
$+(\cdot, \cdot)$	function	2	the function that given two numbers returns the number corresponding to the sum of the two
$<(\cdot, \cdot)$	predicate	2	the "less than" relation between natural numbers

* A constant can be considered as a function with arity equal to 0

Non logical symbols - Example

Example (Domain of arithmetics - extended)

The basic language of arithmetics can be extended with further symbols e.g:

symbols	type	arity	intended meaning
0	constant	0	the smallest natural number
$\text{succ}(\cdot)$	function	1	the function that given a number returns its successor
$+(\cdot, \cdot)$	function	2	the function that given two numbers returns the number corresponding to the sum of the two
$\cdot(\cdot, \cdot)$	function	2	the function that given two numbers returns the number corresponding to the product of the two
$<(\cdot, \cdot)$	predicate	2	the "less than" predicate between natural numbers
$\leq(\cdot, \cdot)$	predicate	2	the "less than or equal to" predicate between natural numbers

Non logical symbols - Example

Example (Domain of strings)

symbols	type	arity	intended meaning
ϵ	constant	0	the empty string
"a", "b",	constants	0	the strings containing one single character of the latin alphabet
$concat(\cdot, \cdot)$	function	2	the function that given two strings returns the string which is the concatenation of the two
$subst(\cdot, \cdot, \cdot)$	function	3	the function that replaces all the occurrence of a string with another string in a third one
$<$	predicate	2	alphabetic order on the strings
$substring(\cdot, \cdot)$	predicate	2	the relation that states if a string is contained in another string

Terms and formulas of FOL

In the following, we write g/n to indicate that the function or predicate symbol g has arity n .

Terms

- every constant c and every variable x is a term;
- if $t_1, \dots, t_n$ are terms and f/n is a function symbol, then $f(t_1, \dots, t_n)$ is a term.

Well formed formulas

- if t_1 and t_2 are terms then $t_1 = t_2$ is a formula;
- If $t_1, \dots, t_n$ are terms and P/n is a predicate symbol, then $P(t_1, \dots, t_n)$ is a formula (if $n = 0$, then the formula is written simply as P);
- if A and B are formulas then $\perp, \top, A \wedge B, A \rightarrow B, A \vee B, \neg A, A \equiv B, (A)$ are formulas;
- if A is a formula and x a variable, then $\forall x.A$ and $\exists x.A$ are formulas (also written simply as $\forall x A$ and $\exists x A$).

Examples of terms and formulas

In the following examples we use the function symbols $f_1/2, f_2/3, g/2, h/3$, the variables x, y, z , the constants a, b, c and the predicate symbols $P/1, A/1, B/1, Q/2$.

Example (terms)

- x
- c
- $f_1(x, c)$
- $f_2(g(x, y), h(x, y, z), y)$

Example (formulas)

- $f_1(a, b) = c$
- $P(c)$
- $\exists x(A(x) \vee B(y))$
- $P(x) \rightarrow \exists y.Q(x, y)$

Semantics of FOL

FOL interpretation

A **first order interpretation** for the alphabet $\langle c_1, c_2, \dots, f_1, f_2, \dots, P_1, P_2, \dots \rangle$ is a pair $\langle \Delta, \mathcal{I} \rangle$ where

- Δ is a non empty set called **interpretation domain**
- $\mathcal{I}$ is a function, called **interpretation function** such that
 - ▶ $\mathcal{I}(c_i) \in \Delta$ for each constant c_i
 - ▶ $\mathcal{I}(f_i) : \Delta^n \rightarrow \Delta$ for each function symbol f_i/n
 - ▶ $\mathcal{I}(P_i) \subseteq \Delta^n$ for each predicate symbol P_i/n

In other words, $\mathcal{I}$ associates to each constant an element of the domain Δ , to each function symbol f_i/n a total n -ary function on the domain Δ , and to each predicate symbol P_i an n -ary relation on the domain Δ .

In the following, we sometime use $c^{\mathcal{I}}$, $f^{\mathcal{I}}$ and $P^{\mathcal{I}}$ instead of $\mathcal{I}(c)$, $\mathcal{I}(f)$ and $\mathcal{I}(P)$, respectively.

Example of interpretation

Example (of interpretation)

Symbols	Constants: <i>alice</i> , <i>bob</i> , <i>carol</i> , <i>robert</i> Function symbol: <i>mother-of</i> /1 Predicate symbol: <i>friends</i> /2
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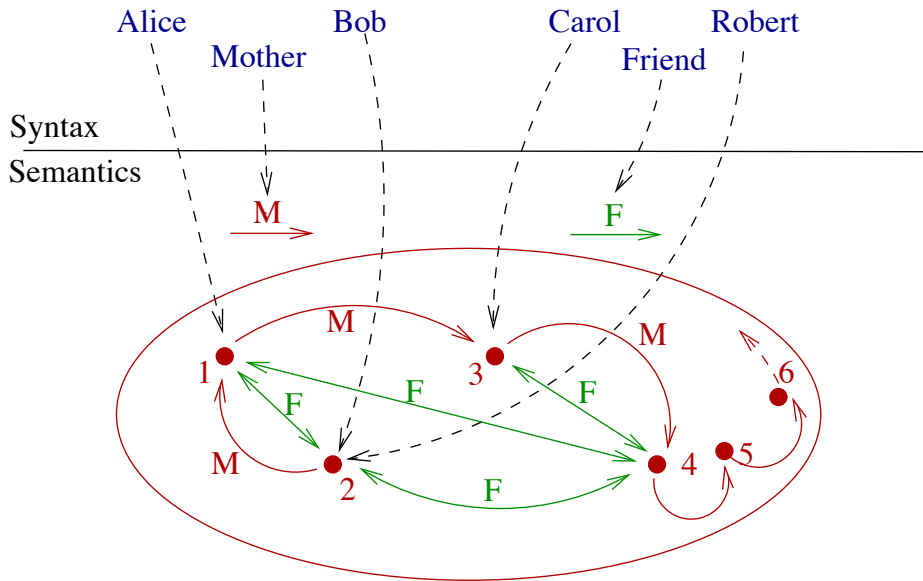
Domain	$\Delta = \{1, 2, 3, 4, \dots\}$
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Interpretation	$\mathcal{I}(\textit{alice}) = 1$, $\mathcal{I}(\textit{bob}) = 2$, $\mathcal{I}(\textit{carol}) = 3$, $\mathcal{I}(\textit{robert}) = 2$
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$\mathcal{I}(\textit{mother-of}) = M$	$M(1) = 3$ $M(2) = 1$ $M(3) = 4$ $M(n) = n + 1$ for $n \geq 4$
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$\mathcal{I}(\textit{friends}) = F =$	$\left\{ \begin{array}{lll} \langle 1, 2 \rangle, & \langle 2, 1 \rangle, & \langle 3, 4 \rangle, \\ \langle 4, 3 \rangle, & \langle 4, 2 \rangle, & \langle 2, 4 \rangle, \\ \langle 4, 1 \rangle, & \langle 1, 4 \rangle, & \langle 4, 4 \rangle \end{array} \right\}$
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Example (cont'd)



Interpretation of terms

Definition (Assignment)

An **assignment** α for $\mathcal{I}$ is a function from the set of variables to Δ .

If α is an assignment for $\mathcal{I}$, then $\alpha[x/d]$ denotes the assignment for $\mathcal{I}$ that coincides with α on all the variables but x , which is associated to d .

Definition (Interpretation of terms)

The **interpretation** of a term t w.r.t. the assignment α , in symbols $t^{\mathcal{I},\alpha}$ is recursively defined as follows (where x is a variable, c a constant and f/n a function symbol):

$$\begin{aligned}x^{\mathcal{I},\alpha} &= \alpha(x) \\c^{\mathcal{I},\alpha} &= \mathcal{I}(c) \\f(t_1, \dots, t_n)^{\mathcal{I},\alpha} &= \mathcal{I}(f)(t_1^{\mathcal{I},\alpha}, \dots, t_n^{\mathcal{I},\alpha})\end{aligned}$$

FOL Satisfaction of formulas

Definition (Satisfaction of a formula by $\mathcal{I}$ w.r.t. α)

The following rules establish when an interpretation $\mathcal{I}$ **satisfies** a formula ϕ w.r.t. the assignment α (written as $(\mathcal{I}, \alpha) \models \phi$):

$(\mathcal{I}, \alpha) \models \top$	and	$(\mathcal{I}, \alpha) \not\models \perp$
$(\mathcal{I}, \alpha) \models t_1 = t_2$	if	$t_1^{\mathcal{I}, \alpha}$ is the same as $t_2^{\mathcal{I}, \alpha}$
$(\mathcal{I}, \alpha) \models P(t_1, \dots, t_n)$	if	$\langle t_1^{\mathcal{I}, \alpha}, \dots, t_n^{\mathcal{I}, \alpha} \rangle \in \mathcal{I}(P)$
$(\mathcal{I}, \alpha) \models \phi \wedge \psi$	if	$(\mathcal{I}, \alpha) \models \phi$ and $(\mathcal{I}, \alpha) \models \psi$
$(\mathcal{I}, \alpha) \models \phi \vee \psi$	if	$(\mathcal{I}, \alpha) \models \phi$ or $(\mathcal{I}, \alpha) \models \psi$
$(\mathcal{I}, \alpha) \models \phi \rightarrow \psi$	if	$(\mathcal{I}, \alpha) \not\models \phi$ or $(\mathcal{I}, \alpha) \models \psi$
$(\mathcal{I}, \alpha) \models \neg \phi$	if	$(\mathcal{I}, \alpha) \not\models \phi$
$(\mathcal{I}, \alpha) \models \phi \equiv \psi$	if	$(\mathcal{I}, \alpha) \models \phi$ if and only if $(\mathcal{I}, \alpha) \models \psi$
$(\mathcal{I}, \alpha) \models (\phi)$	if	$(\mathcal{I}, \alpha) \models \phi$
$(\mathcal{I}, \alpha) \models \exists x \phi$	if	$(\mathcal{I}, \alpha[x/d]) \models \phi$ for some $d \in \Delta$
$(\mathcal{I}, \alpha) \models \forall x \phi$	if	$(\mathcal{I}, \alpha[x/d]) \models \phi$ for all $d \in \Delta$

Example (cont'd)

Exercise

Let α be such that $\alpha(x) = 2$, and let $\mathcal{I}$ be the interpretation defined few slides ago. Check whether the following are true:

- ① $(\mathcal{I}, \alpha) \models \text{Alice} = \text{Bob}$
- ② $(\mathcal{I}, \alpha) \models \text{Robert} = \text{Bob} \wedge \text{friends}(\text{Carol}, \text{mother}(\text{Carol}))$
- ③ $(\mathcal{I}, \alpha[x/2]) \models \neg(x = \text{Bob})$
- ④ $(\mathcal{I}, \alpha) \models \exists x \neg \text{friends}(x, \text{mother}(x))$
- ⑤ $(\mathcal{I}, \alpha) \models \forall x \exists y \text{friends}(x, y)$
- ⑥ $(\mathcal{I}, \alpha) \models \forall x \exists y \neg \text{friends}(x, y)$
- ⑦ $(\mathcal{I}, \alpha) \models \forall x \forall y \text{friends}(x, y) \rightarrow \text{friends}(y, x)$

Free variables

A **free occurrence** of a variable x is an occurrence of x which is not bound to a (universal or existential) quantifier.

Definition (Inductive definition of free occurrence)

- any occurrence of x in t_k is **free** in $P(t_1, \dots, t_k, \dots, t_n)$
- any free occurrence of x in ϕ or in ψ is also **free** in $\phi \wedge \psi$, $\psi \vee \phi$, $\psi \rightarrow \phi$, $\psi \equiv \phi$ and $\neg\phi$
- any free occurrence of x in ϕ , is **free** in $\forall y.\phi$ and $\exists y.\phi$ if y is distinct from x .

Free variables represent individuals which must be instantiated to make the formula a meaningful proposition.

Definition (Free variable)

A **variable x is free** in ϕ (denote by $\phi(x)$) if there is at least a free occurrence of x in ϕ .

Free variables - examples

Intuitively..

Free variables represents individuals which must be instantiated to make the formula a meaningful proposition.

- $friends(Alice, x)$ x free
- $\forall x (friends(Alice, x) \wedge friends(Bob, x))$ x not free
- $(\forall x friends(Alice, x)) \wedge friends(Bob, x)$ x free
- $\forall y. friends(Bob, y)$ no free variables
- $sum(x, 3) = 12$ x free
- $\exists x. (sum(x, 3) = 12)$ no free variables
- $\exists x. (sum(x, y) = 12)$ y free

Open and closed formulas

Definition (Ground/Closed/Open Formula)

A formula ϕ is **ground** if it does not contain any variable.

A formula is **open** if it contains at least one free variable, **closed** otherwise.

Obviously, all ground formulas are closed.

It is not difficult to see that when we interpret a closed formula in an interpretation, we do not actually need any assignment. Indeed, it can be shown that if α_1 and α_2 coincide on the variables that are free in ϕ , then $(\mathcal{I}, \alpha_1) \models \phi$ if and only if $(\mathcal{I}, \alpha_2) \models \phi$. For this reason, in the following, whenever ϕ is a closed formula, we write

$$\mathcal{I} \models \phi$$

to mean that $\mathcal{I}$ satisfies ϕ (independently from any assignment).

An example of representation in FOL

Example (Language)

constants	functions/arity	Predicate/arity
Aldo Bruno Carlo Math DataBase 0, 1, ..., 10	mark/2 best-friend/1	attend/2 friend/2 student/1 course/1 less-than/2

Example (Terms)

Intended meaning	term
an individual named Aldo	Aldo
the mark 1	1
Bruno's best friend	best-friend(Bruno)
anything	x
Bruno's mark in Math	mark(Bruno,Math)
somebody's mark in DataBase	mark(x,DataBase)
Bruno's best friend mark in Math	mark(best-friend(Bruno),Math)